# What Bars Cannot See: Order Flow, Fee-Free Volume, and OHLCV Microstructure Proxies

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Research summary, November 2026 Paper and code: github.com/a0x0h/what-bars-cannot-see

**Question.** Bulk volume classification (BVC), Kyle's lambda, the Amihud ratio, VPIN and high-low spread estimators are computed from price-and-volume bars whenever trade-level data are unavailable. Do they measure order flow and liquidity, can they tell liquidity from volume, and does the part of the flow they miss matter for prices?

**Data and identification.** Binance reports the taker-buy volume of every bar, so the true signed order flow is observed. I use hourly and one-minute bars for twenty coins, January 2022 to October 2026 (about 42,000 hours and 2.5 million minutes per coin), plus a natural experiment: from 14:00 UTC on 8 July 2022 to 00:00 UTC on 22 March 2023 Binance charged no fees on bitcoin pairs, while the other coins kept their fees. The switch hours are visible in the data (Figure 1).

**Methods.** (1) Proxies scored against observed flow and one-minute realised volatility, and a nonparametric *information ceiling*: the $R^2$ of the best possible function of the bar's own return, an upper bound for every return-based classifier. (2) Event-window difference-in-differences of BTC/USDT against nineteen control coins around both fee switches, with permutation tests across coins and over time (one treated unit), and trade-level data around the switches. (3) Return response to flow shocks with Newey-West and Driscoll-Kraay inference, and a purged monthly walk-forward (elastic net, boosting, Clark-West tests, trading costs, deflated Sharpe ratio).

**Findings.**

1. *Bars see volatility, not order flow.* Range-based volatility has a correlation of at least 0.93 with realised volatility in every coin. But in nineteen of twenty coins no function of the hourly bar's return explains more than about a quarter of the variance of aggressor imbalance (0.25 for bitcoin, less for thinly traded coins), and BVC sits close to this bound. Spread estimators measure volatility (Corwin-Schultz: 15.8 bps on a 0.002-bps tick).
2. *Volume is not liquidity.* Removing fees multiplied BTC/USDT trades by 5.3 relative to the controls, with breaks in the exact hours of both switches. 93% of the extra volume was offset by opposite aggressor flow within the same minute, and the share offset within the same second doubled (30% to 59%). The Amihud ratio fell by 69% and BVC-based Kyle's lambda by 33%, as if liquidity had improved sharply, while the price impact of a dollar of observed net flow showed no significant change and order-book studies find wider spreads. The same balanced volume then moved to BTC/TUSD and BTC/FDUSD as the promotion moved. Volume-based measures cannot tell volume from liquidity.
3. *The hidden flow is the informative flow.* In bitcoin a one-SD imbalance shock is followed by a 1.3 bp reversal over two hours ($t = -4.0$), unchanged when the bar's return is held fixed; hourly BVC predicts nothing. Across twenty coins the pooled response is $-0.65$ bp ($t = -3.0$) and is driven by market-wide flow. It is detectable out of sample (Clark-West $p = 0.0001$) but breaks even at about 1.2 bp, and no multivariate model beats a zero forecast over 53 test months.

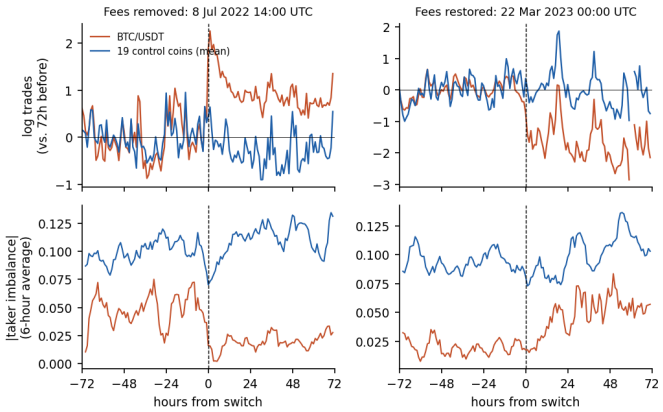



**Figure 1:** BTC/USDT against the average of nineteen control coins in the 72 hours around the removal (left) and restoration (right) of fees: log trades relative to the pre-switch mean (top) and absolute taker imbalance (bottom).

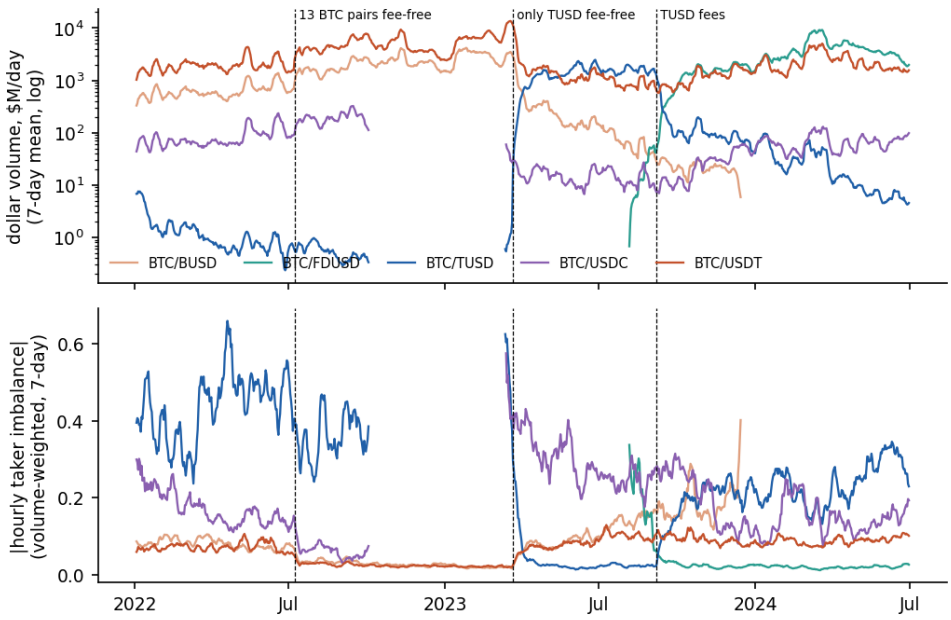



**Figure 2:** The volume follows the fee. Daily dollar volume (top) and absolute hourly taker imbalance (bottom) of bitcoin against five stablecoins. High volume with almost perfectly balanced aggressor flow appears on BTC/USDT and BTC/BUSD while all bitcoin pairs are free, on BTC/TUSD once it is the only free pair, and on BTC/FDUSD after that.

**Why it matters.** Much empirical work on cryptocurrency and other markets has only bars. These results show that bar-based flow measures miss most of the flow, that volume-based liquidity measures are not comparable across fee regimes or incentive schemes that generate two-sided trading, and that the flow bars miss is the part that moves prices. The natural next steps are order-book data at higher frequency, flow across venues, and identifying who trades fee-free two-sided volume.

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